AF-400 INVERTER DRIVES

INVERTER ASSEMBLY (200 KVA) - 331X402AAG01

OPERATION - TROUBLE SHOOTING - REPAIR

GEK-62518

DIAGRAMS

INVERTER ASM	ELEM	36C764880AA
	CONN	36D868886AA
PHASE MODULE	ELEM	36C764878AA
	CONN	36C764882AA
CONVERTER MODULE	ELEM	36C764879AA
	CONN	36C764887AA
COMM. POWER SUPPLY	ELEM	36C764881AA
	CONN	36D868890AA

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC

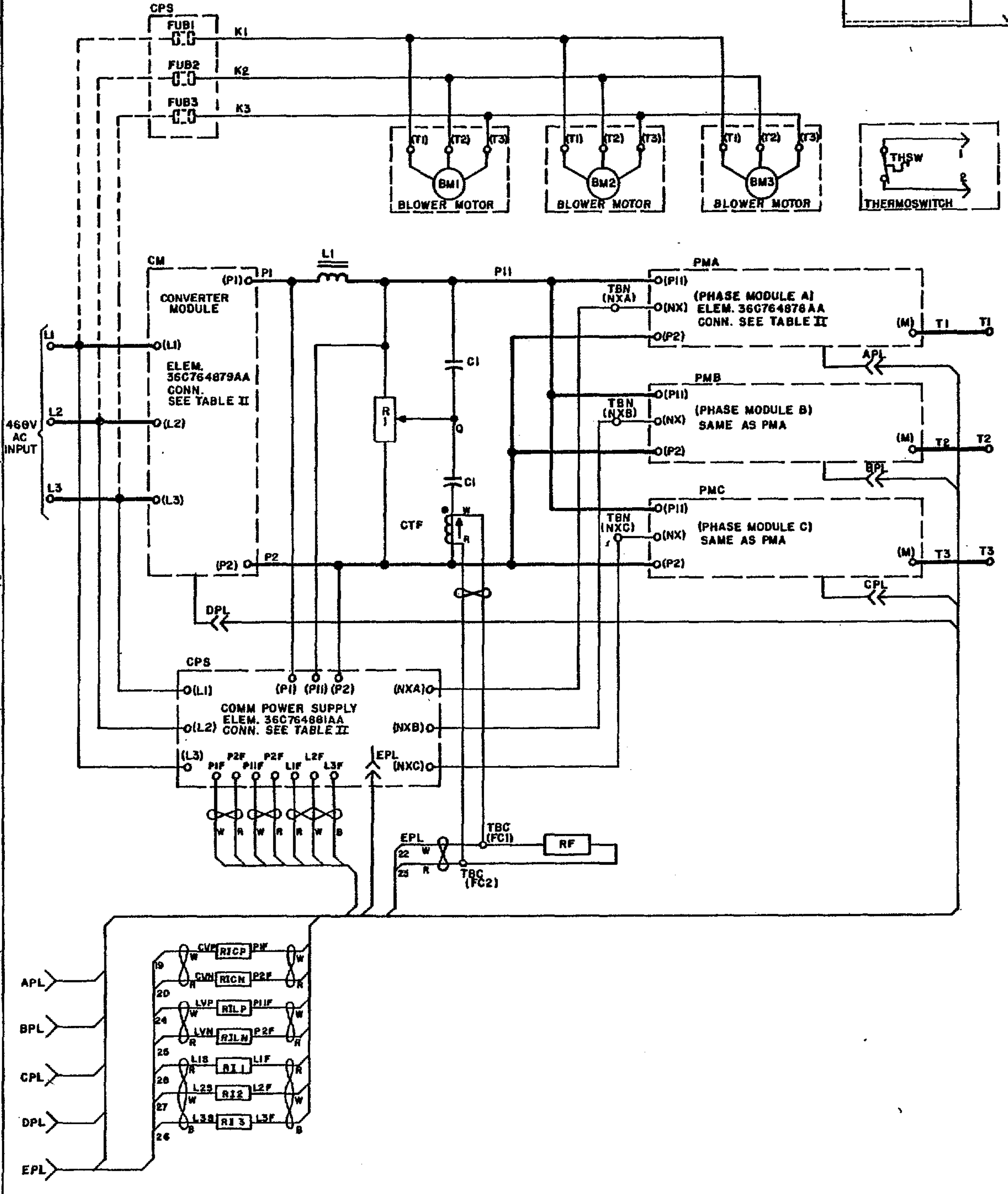
[illegible] of the General Electric Co

GENERAL ELECTRIC

36 4880AA	CONT ON SHEET · SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:–				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	 DECIMALS	 ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-400 INVERTER ASM!
FIRST MADE FOR AF-400
331X401, 402, 403AA


CPS
FUB1
FUB2
FUB3
K1
K2
K3
(T1)
(T2)
(T3)
BM1
BM2
BM3
BLOWER MOTOR
BLOWER MOTOR
BLOWER MOTOR
THSW
THERMOSWITCH
CM
CONVERTER MODULE
ELEM. 36C764879AA CONN. SEE TABLE II
(P1)
P1
(L1)
(L2)
(L3)
(P2)
P2
L1
L2
L3
460V AC INPUT
L1
P11
C1
C1
R1
Q
CTF
W
R
PMA
(PHASE MODULE A) ELEM. 36C764876AA CONN. SEE TABLE II
PMB
(PHASE MODULE B) SAME AS PMA
PMC
(PHASE MODULE C) SAME AS PMA
TBN (NXA)
TBN (NXB)
TBN (NXC)
(P11)
(NX)
(P2)
(M)
T1
T2
T3
APL
BPL
CPL
DPL
EPL
CPS
COMM POWER SUPPLY ELEM. 36C764881AA CONN. SEE TABLE II
(P1) (P11) (P2)
(NXA)
(NXB)
(NXC)
P1F P2F P11F P2F L1F L2F L3F
TBC (FC1)
TBC (FC2)
RF
22
23
CVP
CVN
LVP
LVN
L1S
L2S
L3S
R1CP
R1CN
R1LP
R1LN
R1 1
R1 2
R1 3
P1F
P2F
P11F
P2F
L1F
L2F
L3F
19
20
24
25
26
27
24
W
R
B
A
B
C
D
E
1
2
3
4
5
6

} OVER TEMP. IND.

NOTES:

1. *THE* ASSEMBLY CATALOG NUMBER GROUP DISTINCTION, MODULE CATALOG NUMBERS AND SOME COMPONENT VALUES ARE DEPENDENT UPON KVA RATING. TABLE I LISTS VARIABLE COMPONENT VALUES.
2. THE L1 CHOKE IS MOUNTED *REMOTE TO THE INVERTER ASSEMBLY.* ITS VALUE IS DEPENDENT UPON KVA RATING AND IS LISTED IN TABLE I.
3. *ELEMENTARY* WIRING SYMBOLS
   - W R —INDICATES TWISTED WIRES, COLOR INDICATED BY LETTER.
   - 1 2 3 —INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG.
   - →≻—QUICK DISCONNECT "STAB–ON" CONNECTION.
   - —DOT INDICATES POSITIVE TERMINAL *WHEN CURRENT FLOWS* IN THE DIRECTION OF THE ARROW.
4. *REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION* OF THIS ASSEMBLY.
5. RESISTORS RIC, RIL, RI1, RI2, RI3 ARE, RESISTANCE ISOLATOR RESISTORS, AND ARE *MOUNTED AS AN INTEGRAL PART OF THE WIRE HARNESS.*
6. BM1, BM2, BM3 USED ON 300/400 KVA
   BM1, BM2 USED ON 200 KVA
   BM1 USED ON 100/50 KVA
7. SEE TABLE II FOR CONNECTION DIAGRAMS.
8. TABLE I

KVA	331X	L1	C1//C1	C1	R1	RF
		μH	μF	μF	OHM/W	OHM
400	401AAG01	400	22400	44800	1000/800	8.3 4X33.2Ω IN PAR.
300	401AAG02	530	16800	33600	1500/400	11.1 3X33.2Ω IN PAR.
200	402AAG01	800	11200	22400	1500/400	16.6 2X33.2Ω IN PAR.
100	403AAG01	1600	5600	11200	2500/200	16.6 2X33.2Ω IN PAR
50	403AAG02	3200	2800	5600	2500/200	33.2Ω

TABLE II

KVA	PHASE MODULE	CONVERTER MODULE	INVERTER ASSEMBLY	COMM. P/S
400/300	36C764882AA	36C764887AA	36D868885AA	36D868889AA
200/	36C764883AA	36C764888AA	36D868886AA	36D868890AA
100/50	36C764884AA	36C764889AA	36D868887AA	36D868891AA

REV NO.
Ø
1

DRAWING NO.
CONT ON SHEET · SH NO.

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1.	May 20, 1999 J. Binder	AB	

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY md Pyh AUG. 17, 1976
ISSUED O.E. Williams Jan 31, 1977
APPROVALS SBS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764880AA
CONT ON SHEET · SH NO.

FF-809-F (9-75)
PRINTED IN U.S.A.

GENERAL ELECTRIC 36D868886AA

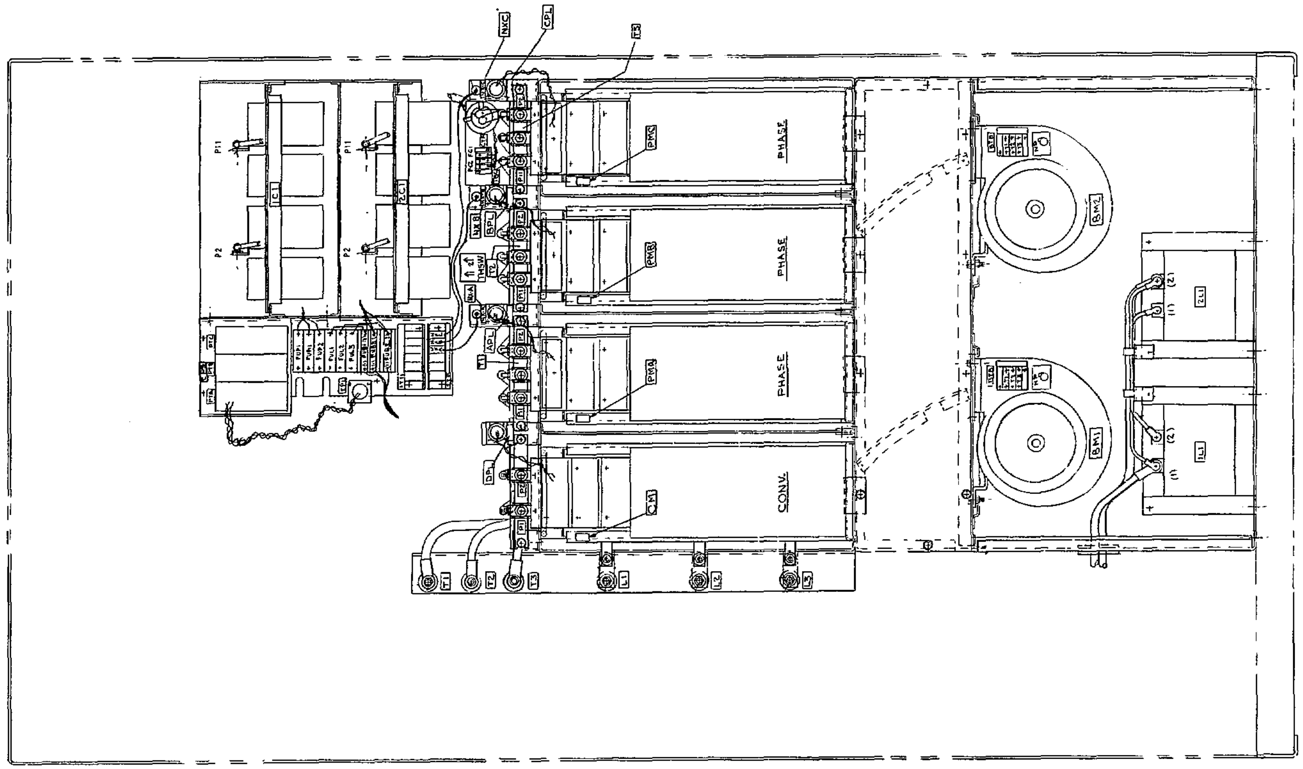
TITLE INVERTER CONNECTION DIAG

FIRST MADE FOR AF400 FOR 200KVA

REV NO Ø1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES.	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±



INVERTER ASM WIRING - 200 KVA

WIRE NO.	CONNECTION
	BUS BAR - 1/4" X 1"
L1	PTB(L1), CH(L1)
L2	PTB(L2), CH(L2)
L3	PTB(L3), CM(L3)
	#2/0 BLK-SILICONE
P1	1L1(1), CH(P1)
P2	CH(P2), PHA(P2), PHB(P2), PHC(P2)
P11	2L1(2), PMA(P11), PHB(P11), PHC(P11)
T1	PTB(T1), PHA(H)
T2	PTB(T2), PHB(H)
T3	PTB(T3), PHC(H)
	#4 BLK-SILICONE
P1	1L1(1), 2L1(1)
P11	1L1(2), 2L1(2)
	#6 BLK-SILICONE
P2	1C1(P2), PMC(P2)
P2	2C1(P2), PMC(P2)
P11	1C1(P11), PHC(P11)
P11	2C1(P11), PHC(P11)
	#14 BLK
L1	BUS(L1), FUB1(2)
L2	BUS(L2), FUB2(2)
L3	BUS(L3), FUB3(2)
K1	FUB1(1), BH1(T1), BH2(T1)
K2	FUB2(1), BH1(T2), BH2(T2)
K3	FUB3(1), BH1(T3), BH2(T3)
NXA	2TB(NXA), TBN(NXA)
NXB	2TB(NXB), TBN(NXB)
NXC	2TB(NXC), TBN(NXC)
P1	FUP1(2), CH(P1)
P2	FUP2(2), CH(P2)
P11	FUP11(2), PHA(P11)
	RESISTOR WIRES
P1F W	1TB(P1F), EPL(19)
P2F R	1TB(P2F), EPL(20)
P11F W	1TB(P11F), EPL(24)
P2F R	1TB(P2F), EPL(25)
L1F R	1TB(L1F), EPL(28)
L2F W	1TB(L2F), EPL(27)
L3F B	1TB(L3F), EPL(26)
	PLUG WIRING
APL	DRIVER(APL), PHA(APL)
BPL	DRIVER(BPL), PHB(BPL)
CPL	DRIVER(CPL), PHC(CPL)
DPL	DRIVER(DPL), CH(DPL)
EPL	DRIVER(EPL), CPS(EPL)
EPL22 W	DRIVER(EPL), TBC(FC1)
EPL23 R	DRIVER(EPL), TBC(FC2)

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

REVISIONS

1 JAN 30, 1978 SV B12 #21
CHG RESISTOR WIRES IN WIRE TABLE

PRINTS TO	
SB(8)M	SJ(BW)
SR(2)BW	
SL(2)BW	
AW(BW)	
SD(CD)	
JA(CD)	

Dec 30, 1976
Williams Jan 31 1977

SPEED VARIATOR
ERIE PA.

36D868886AA

FF-812-F(9-75) PRINTED IN U.S.A.

GENERAL ELECTRIC

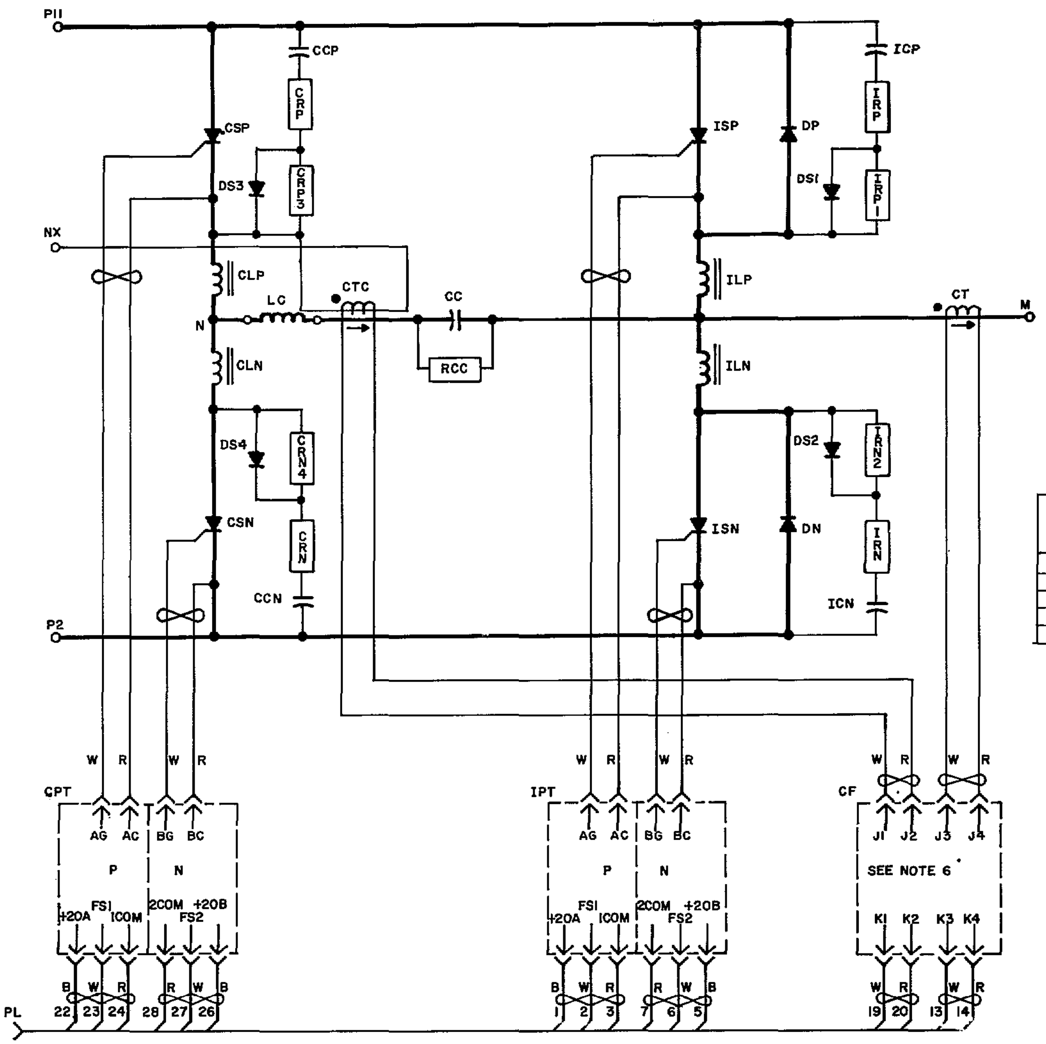
36C764878AA

CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE ELEMENTARY DIAGRAM
AF-400 PHASE MODULE

FIRST MADE FOR AF-400
331X411, 412, 413AA



NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES OF I OR C TO INDICATE INVERTER OR COMMUTATION SECTION. SUFFIXES OF P OR N INDICATE POSITIVE OR NEGATIVE. THE BASIC NOMENCLATURE IS:
   - C — CAPACITOR
   - D — DIODE
   - L — CHOKE
   - PT — PULSE TRANSFORMER PRINTED CIRCUIT CARD
   - PL — CONNECTOR PLUG
   - R — RESISTOR
   - S — SCR
   - DS — SNUBBER DIODE
2. PULSE TRANSFORMER CARD CIRCUITRY IS NOT SHOWN FOR SIMPLICATION. FOR DETAILS REFER TO THE INSTRUCTION BOOK.
3. ELEMENTARY WIRING SYMBOLS
   - R W B —INDICATES TWISTED WIRES, COLOR INDICATED BY LETTER.
   - 1 2 3 —INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG.
   - —QUICK DISCONNECT "STAB-ON" CONNECTION.
   - —DOT INDICATES POSITIVE TERMINAL WHEN CURRENT FLOWS IN THE DIRECTION OF THE ARROW.
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
5.

KVA	331X	CC	LC	IRP, CRP IRN, CRN	IRP1, CRP3 IRN2, CRN4	ICP, CCP ICN, CCN	ILP ILN	CLP CLN	RCC
		μF	μH	OHM/W	OHM/W	μF	SAT μH	SAT μH	K OHM /W
400	411AAG01	80	5	2.5/40	20/40	1	1.5	1.5	280/4
300	411AAG02	60	7.5	2.5/40	20/40	1	2	2	280/4
200	412AAG01	40	10	2.5/40	20/40	1	2.6	2.6	560/2
100	413AAG01	20	25	10/40	20/40	.25	4	4	560/2
50	413AAG02	10	53	10/40	20/40	.25	8	8	1000/2

6. THE CURRENT FEEDBACK CARD WILL CHANGE DEPENDING ON KVA RATING. GROUP OF CARD AND JUMPER SELECTION FOR NOMINAL CURRENT ARE—

KVA	CF 193X382AA	NOMINAL INV. CURRENT A RMS	JUMPER SELECTION	NOMINAL COMM. CURRENT A PEAK
400	G05	580 500 420	XC XB XA	2100
300	G04	420 350 280	XC XB XA	1500
200	G03	280 210 175	XC XB XA	1000
100	G02	140 105 84	XC XB XA	500
50	G01	70 56 42	XC XB XA	250

REV NO.
1

DRAWING NO.
CONT ON SHEET SH NO.

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	May 20, 19?? J. Burden	JAS	JHB

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY M.B.Pof— AUG. 16, 1976
ISSUED W. E. Williams Jan. 31 1977
APPROVALS EBS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764878AA
CONT ON SHEET SH NO.

FF-809-F (9-75)
PRINTED IN U.S.A.

GENERAL ELECTRIC

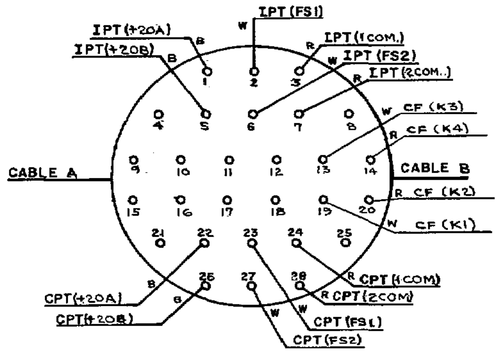
36C764882AA
CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE **CONNECTION DIAGRAM**
400/300 KVA PHASE MODULE
FIRST MADE FOR **AF-400**
331X411AAG01

REF TO WIRE CHART
36A3C8012XBG01

VIEW FROM WIRE SIDE OF PLUG SOCKET


IPT (FS1)
IPT(+20A)
IPT(+20B)
IPT(1COM.)
IPT(FS2)
IPT(2COM.)
W
B
R
1
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3
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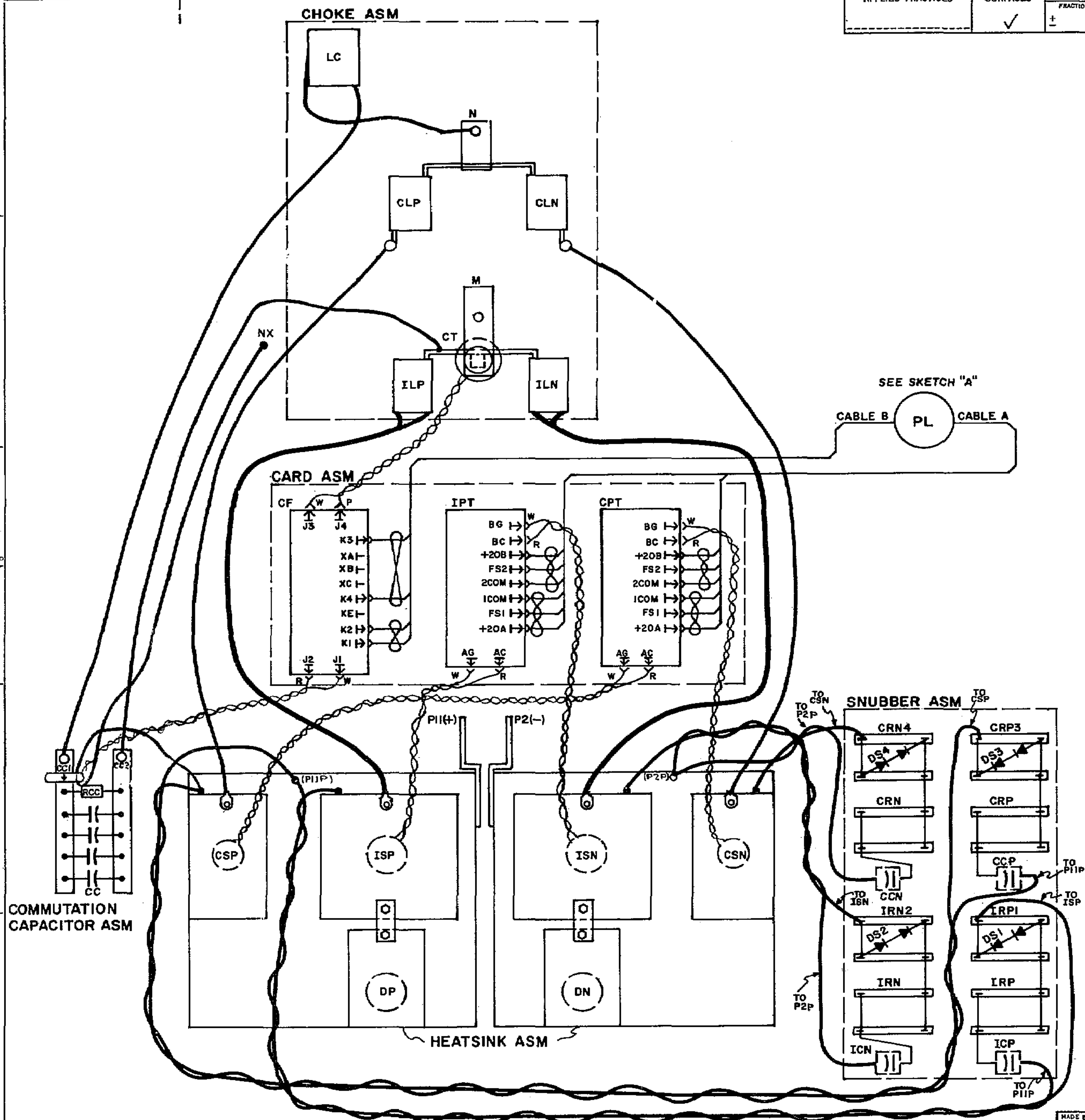
CF (K3)
CF (K4)
CABLE A
CABLE B
CF (K2)
CF (K1)
CPT(1COM)
CPT(2COM)
CPT(+20A)
CPT(+20B)
CPT(FS1)
CPT(FS2)

SKETCH "A"


CHOKE ASM
LC
N
CLP
CLN
M
CT
ILP
ILN
NX
SEE SKETCH "A"
CABLE B
PL
CABLE A
CARD ASM
CF
W
P
J3
J4
K3
XA1
XB1
XC1
K4
KE1
K2
K1
J2
J1
R
IPT
CPT
BG
BC
+20B
FS2
2COM
1COM
FS1
+20A
AG
AC
P1(+)
P2(-)
(P1P)
(P2P)
CSP
ISP
ISN
CSN
DP
DN
HEATSINK ASM
COMMUTATION CAPACITOR ASM
CC1
CC2
RCC
CC
SNUBBER ASM
TO CSN
TO P2P
TO CSP
CRN4
DS4
CRN
CCN
TO ISN
IRN2
DS2
IRN
ICN
CRP3
DS3
CRP
CCP
TO P1P
TO ISP
IRP1
DS1
IRP
ICP

NOTES.
1. ALL POWER WIRING IS EITHER BUS OR CONNECTORS PER MECHANICAL ASSEMBLY DRAWING.
2. WIRES ARE MARKED AS INDICATED. AC, AG ; BC & BG ARE TWISTED PAIRS OF #20 AWG RED & WHITE. GATES (G) ARE WHITES.
   TWISTED RED, WHITE AND BLUE WIRES TO PULSE TRANSFORMER CARDS, ARE #20 WIRES
   TWISTED RED & WHITE WIRES TO CURRENT FEEDBACK CARD ARE #20 AWG.
3. ALL OTHER CONTROL WIRES ARE #16 AWG RED, EXCEPT AS OTHERWISE INDICATED.

REV NO.
0
1

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	FEB 25 1997 W. Bunden	RECOR CHG

P2P WAS P2, P1P WAS P1
CHGD CONN. FROM LC — M TO LC — N, CHGD CONN FROM CC2 — N TO CC2 — M
CHGD SK. "A"

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY M B Pople MAY 20, 1976
ISSUED MAY 26, 1976

APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764882AA
CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FF-809-F (9-75)
PRINTED IN U.S.A.

GENERAL ELECTRIC

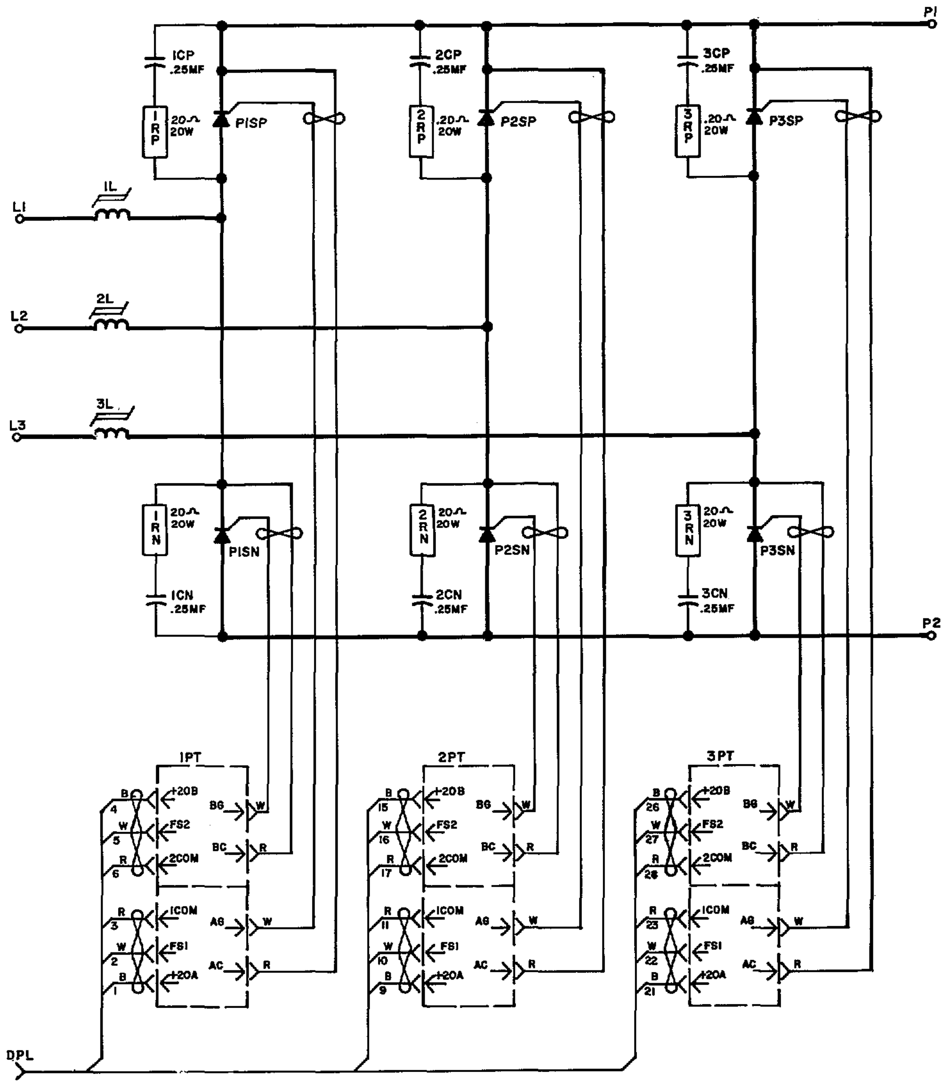
36C764879AA

CONT ON SHEET SH NO.

TITLE ELEMENTARY DIAGRAM
AF-400 CONVERTER MODULE
FIRST MADE FOR STANDARD LINE
331X421, 422, 423AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±



P1
P2
1CP .25MF
2CP .25MF
3CP .25MF
1RP 20Ω 20W
2RP 20Ω 20W
3RP 20Ω 20W
P1SP
P2SP
P3SP
L1
L2
L3
1L
2L
3L
1RN 20Ω 20W
2RN 20Ω 20W
3RN 20Ω 20W
P1SN
P2SN
P3SN
1CN .25MF
2CN .25MF
3CN .25MF
1PT
2PT
3PT
+20B
FS2
2COM
1COM
FS1
+20A
BG
BC
AG
AC
W
R
B
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23
22
21
DPL

NOTES

1. THIS DIAGRAM IS FUNCTIONALLY NOMENCLATED WITH PREFIXES P1, P2 ETC., TO INDICATE PHASE CONTROL AND PHASE. SUFFIXES OF P OR N INDICATE POSITIVE OR NEGATIVE. BASIC NOMENCLATURE IS:
   - C — CAPACITOR
   - D — DIODE
   - L — CHOKE
   - PT — PULSE TRANSFORMER PRINTED CIRCUIT CARD
   - PL — CONNECTOR PLUG
   - R — RESISTOR
   - S — SCR
   - X — AUXILIARY SCR
2. PULSE TRANSFORMER CARD CIRCUITRY IS NOT SHOWN FOR SIMPLIFICATION. FOR DETAILS REFER TO THE INSTRUCTION BOOK.
3. ELEMENTARY WIRING SYMBOLS
   - R W B — INDICATES TWISTED WIRES, COLORS INDICATED BY LETTER
   - 1 2 3 — INDICATES CABLE ASSEMBLY, NUMBERS ARE POINTS IN CONNECTOR PLUG PL.
   - —>>— — QUICK DISCONNECT "STAB ON" CONNECTION
4. REFER TO THE CONNECTION DIAGRAM FOR A PHYSICAL REPRESENTATION OF THIS ASSEMBLY.
5. 1L, 2L, 3L; 3 FERRITE CORES 1 TURN

DRAWING NO.
CONT ON SHEET SH NO.

REV NO. 0

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY MDPozsgai AUG. 24, 1976
ISSUED O E Williams Jr. 31 1977
APPROVALS EASOMS
SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION
36C764879AA
CONT ON SHEET SH NO.

GENERAL ELECTRIC 36C764887AA

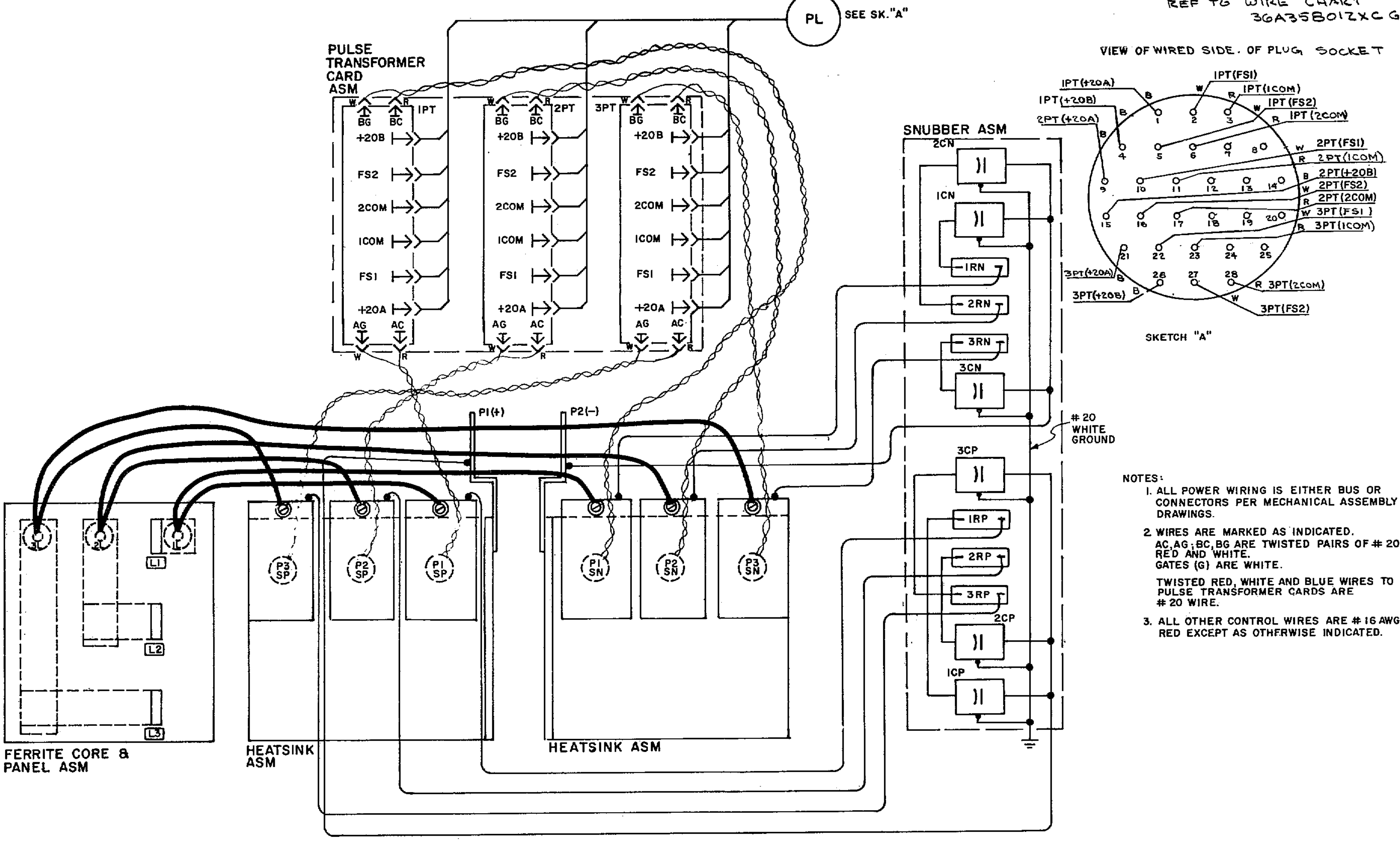
CONT ON SHEET SH NO.

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE CONNECTION DIAGRAM
400 / 300 KVA CONVERTER MODULE

FIRST MADE FOR AF-400
331X421AAG01

REF TO WIRE CHART
36A35B012XC G01


PL
SEE SK."A"
PULSE TRANSFORMER CARD ASM
W
R
BG
BC
1PT
2PT
3PT
+20B
FS2
2COM
1COM
FS1
+20A
AG
AC
P1(+)
P2(−)
P3 SP
P2 SP
P1 SP
P1 SN
P2 SN
P3 SN
L1
L2
L3
FERRITE CORE & PANEL ASM
HEATSINK ASM
HEATSINK ASM
SNUBBER ASM
2CN
1CN
1RN
2RN
3RN
3CN
3CP
1RP
2RP
3RP
2CP
1CP
# 20 WHITE GROUND
VIEW OF WIRED SIDE. OF PLUG SOCKET
1PT(+20A)
1PT(+20B)
2PT(+20A)
1PT(FS1)
1PT(1COM)
1PT(FS2)
1PT(2COM)
2PT(FS1)
2PT(1COM)
2PT(+20B)
2PT(FS2)
2PT(2COM)
3PT(FS1)
3PT(1COM)
3PT(+20A)
3PT(+20B)
3PT(2COM)
3PT(FS2)
B
W
R
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2
3
4
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6
7
8
9
10
11
12
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15
16
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SKETCH "A"

NOTES:

1. ALL POWER WIRING IS EITHER BUS OR CONNECTORS PER MECHANICAL ASSEMBLY DRAWINGS.
2. WIRES ARE MARKED AS INDICATED.
   AC, AG ; BC, BG ARE TWISTED PAIRS OF # 20AWG RED AND WHITE.
   GATES (G) ARE WHITE.

   TWISTED RED, WHITE AND BLUE WIRES TO PULSE TRANSFORMER CARDS ARE # 20 WIRE.
3. ALL OTHER CONTROL WIRES ARE # 16 AWG RED EXCEPT AS OTHERWISE INDICATED.

DRAWING NO. CONT ON SHEET SH NO.

REV NO.
~~0~~ ~~1~~ 2

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	M.B. Pazha	SEPT.22 1976	SV-867*3
2	FEB 25 1977 J. Bender		RECOR CHG
	CHG'D SK A TO WIRE SIDE VIEW		

PRINTS TO	
5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY M B Pazha MAY 12, 1976
ISSUED MAY 26 1976
APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764887AA
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GENERAL ELECTRIC

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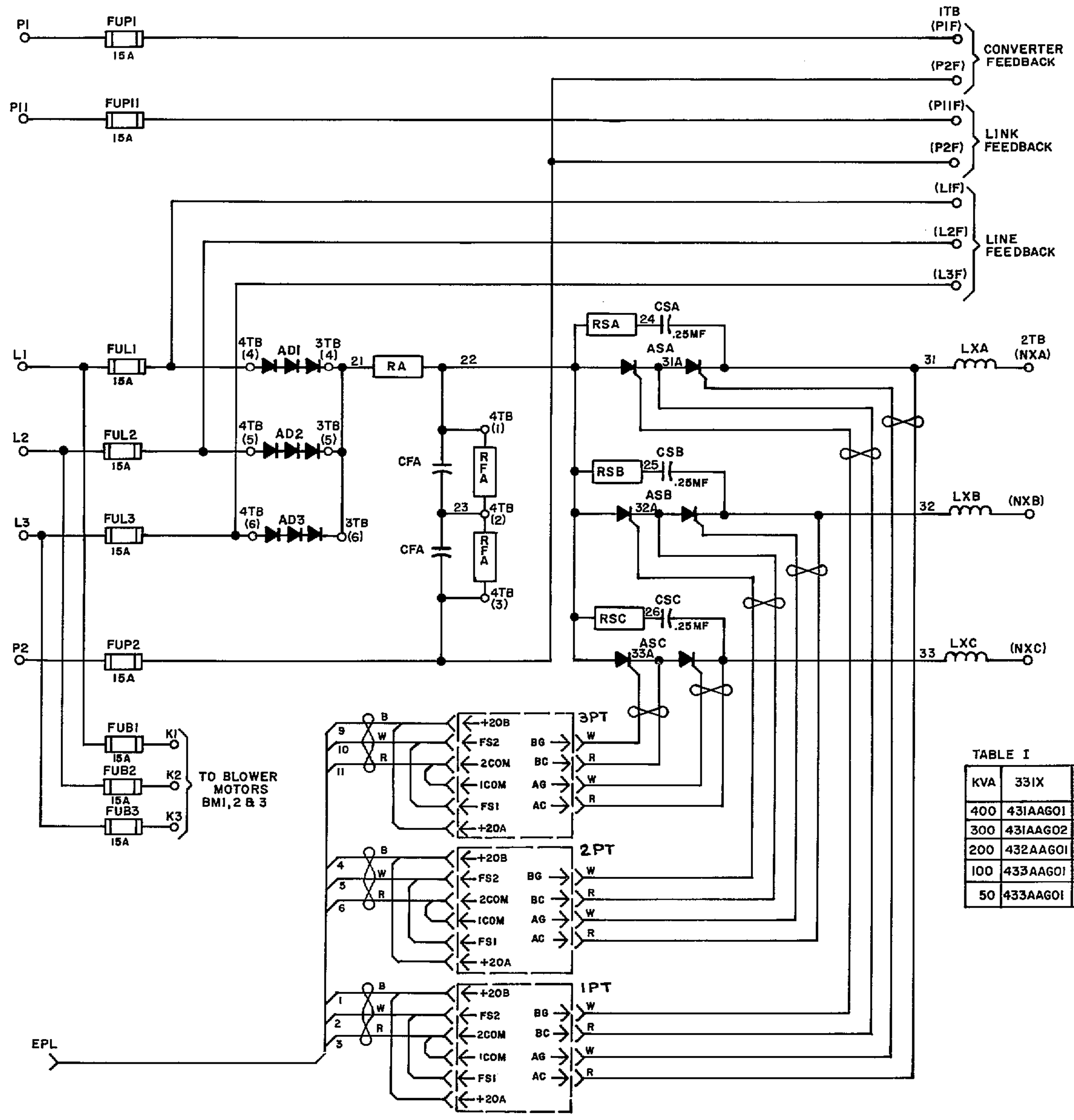
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	 DECIMALS	 ANGLES
	✓	±	±	±

TITLE **ELEMENTARY DIAGRAM**
COMMUTATION POWER SUPPLY

FIRST MADE FOR **AF-400**
331X431AAG01

DRAWING NO.
CONT ON SHEET SH NO.



P1
FUP1
15A
P11
FUP11
15A
1TB
(P1F)
(P2F)
CONVERTER FEEDBACK
(P11F)
(P2F)
LINK FEEDBACK
(L1F)
(L2F)
(L3F)
LINE FEEDBACK
L1
FUL1
15A
L2
FUL2
15A
L3
FUL3
15A
P2
FUP2
15A
4TB (4)
AD1
3TB (4)
4TB (5)
AD2
3TB (5)
4TB (6)
AD3
3TB (6)
21
RA
22
4TB (1)
CFA
RFA
23
4TB (2)
CFA
RFA
4TB (3)
RSA
24
CSA
.25MF
ASA
31A
RSB
25
CSB
.25MF
ASB
32A
RSC
26
CSC
.25MF
ASC
33A
31
LXA
2TB (NXA)
32
LXB
(NXB)
33
LXC
(NXC)
FUB1
15A
K1
FUB2
15A
K2
FUB3
15A
K3
TO BLOWER MOTORS BM1, 2 & 3
3PT
2PT
1PT
B
W
R
9
10
11
4
5
6
1
2
3
+20B
FS2
2COM
1COM
FS1
+20A
BG
BC
AG
AC
EPL

TABLE I

KVA	331X	LX µH	CFA µF	RA OHMS/W	RFA OHM/CONN/W	RSA OHM/CONN/W
400	431AAG01	54	1400	5/100	10K/2X5K SER/20	100/2 X 200 PAR/40
300	431AAG02	54	1400	5/100	"	"
200	432AAG01	110	850	10/50	"	200/2X100 SER/40
100	433AAG01	220	850	10/50	"	400/2X200 SER/40
50	433AAG01	"	850	10/50	"	"

REV NO.
~~0~~ ~~1~~ ~~2~~ ~~3~~ 4

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

1	Feb 1, 1977 E. Gualtieri DW	SV872 #12A	
2	NOV. 7, 1977 G.R. Baker	PER SV-872 #13A	
3	JAN 31, 1978 [illegible]	SV872 #20	CHG MARKING 1PT WAS PTA 2PT WAS PTB 3PT WAS PTC
4	M. Crotty June 21, 19[illegible]	PER SV-872 #8	

PRINTS TO	
5B(8)M	5J(BW)
5R(2)BW	
5L(2)BW	
AW(BW)	
5D(CD)	
JA(CD)	

MADE BY M D Pazha AUG 25, 1976
ISSUED W. Kazanauskas OCT 18, 1976
APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764881AA
CONT ON SHEET SH NO.

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

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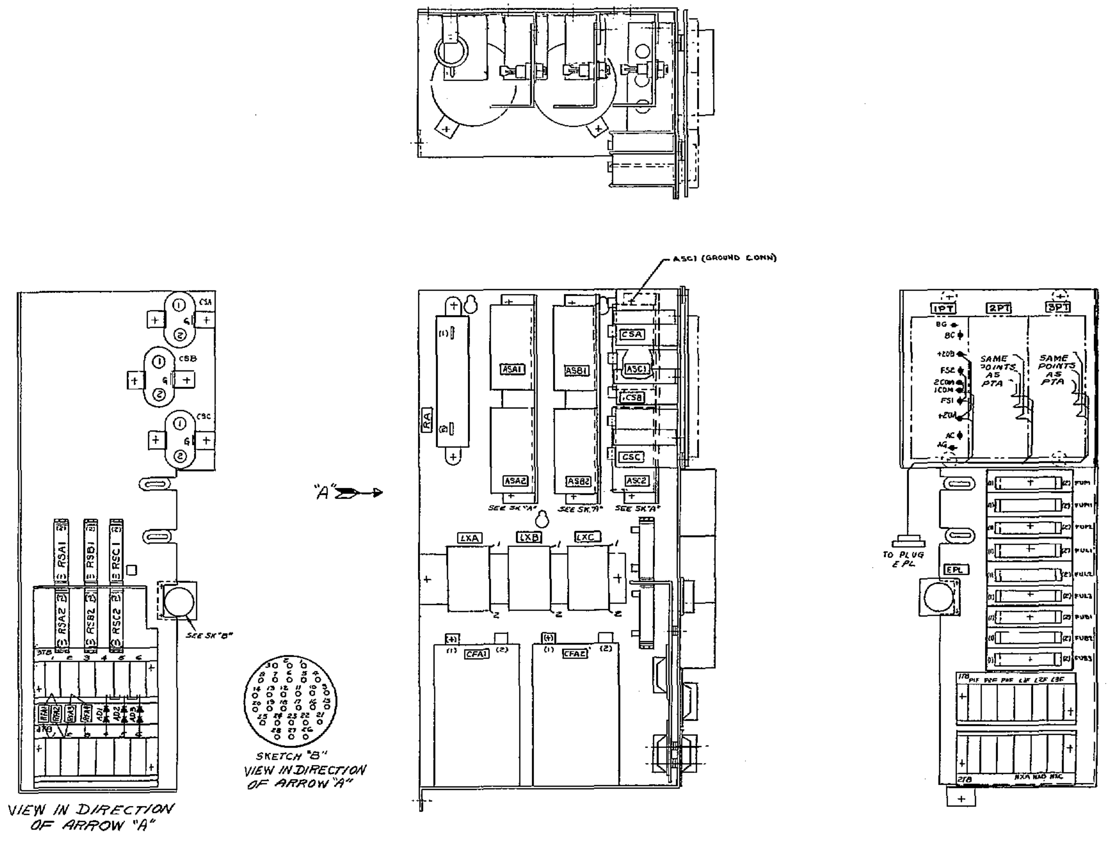
GENERAL ELECTRIC 36D868890AA

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

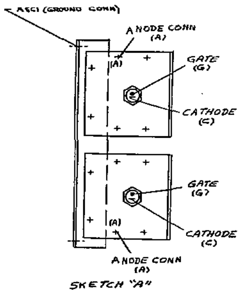
APPLIED PRACTICES-	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

REV. NO. ~~0~~ 1 & 3

TITLE CONNECTION DIAGRAM
COMMUTATION POWER SUPPLY
FIRST MADE FOR AF400
~~300~~ 200 KVA 3S1X432AA G1 ~~G2~~



ASC1 (GROUND CONN)
CSA
CSB
CSC
RA
ASA1
ASA2
ASB1
ASB2
ASC1
ASC2
SEE SK "A"
SEE SK "A"
SEE SK "A"
LXA
LXB
LXC
CFA1
CFA2
"A"
RSA1
RSB1
RSC1
RSA2
RSB2
RSC2
3TB
AD1
AD2
AD3
SEE SK "B"
SKETCH "B"
VIEW IN DIRECTION OF ARROW "A"
VIEW IN DIRECTION OF ARROW "A"
1PT
2PT
3PT
BG
BC
+20B
FS2
2COM
1COM
FS1
+20A
AC
AG
SAME POINTS AS PTA
SAME POINTS AS PTA
FUP1
FUP11
FUP2
FUL1
FUL2
FUL3
FUB1
FUB2
FUB3
1TB
P1F P2F P11F L1F L2F L3F
2TB
NXA NXB NXC
TO PLUG EPL
EPL



ASC1 (GROUND CONN)
ANODE CONN (A)
GATE (G)
CATHODE (C)
GATE (G)
CATHODE (C)
ANODE CONN (A)

SKETCH "A"

	WIRE NO.	CONNECTION
		18 BLACK—POWER WIRING
	P1F	FUP1(1),1TB(P1F)
	P2F	1TB(P2F),FUP2(1),CFA2(2),4TB(3)
	P11F	1TB(P11F),FUP11(1)
	L1F	1TB(L1F),FUL1(1),4TB(4)
	L2F	1TB(L2F),FUL2(1),4TB(5)
	L3F	1TB(L3F),FUL3(1),4TB(6)
	21	3TB(4)*,3TB(5)*,3TB(6)*,RA(2)
	22	4TB(1),CFA1(1),ASC1(A),ASB1(A), ASA1(A),RA(1)
	23	4TB(2),CFA2(1),CFA1(2)
	22	ASA1(A),RSC1(2),RSA1(2),RSB1(2)
	24	CSA(1),RSA2(2)
	24A	RSA1(1),RSA2(1)
	25	CSB(1),RSB2(2)
	25A	RSB1(1),RSB2(1)
	26	CSC(1),RSC2(2)
	26A	RSC1(1),RSC2(1)
1	31	CSA(2),LXA(1),ASA2(C) ▲
	32	CSB(2),LXB(1),ASB2(C) ▲
	33	CSC(2),LXC(1),ASC2(C) ▲
	NXA	2TB(NXA),LXA(2)
	NXB	2TB(NXB),LXB(2)
	NXC	2TB(NXC),LXC(2)
	31A	ASA1(C)▲,ASA2(A)
1	32A	ASB1(C)▲,ASB2(A)
	33A	ASC1(C)▲,ASC2(A)
1	G	CSC(G),CSB(G),CSA(G),ASC1(G)
	L1	FUL1(2),FUB1(2)
	L2	FUL2(2),FUB2(2)
	L3	FUL3(2),FUB3(2)
		* METAL JUMPER CONNECTION
		20 RED & WHITE TWISTED SCR GATE LEAD WIRING
	W	ASA1(G), 1PT (BG)
	R	ASA1(C)▲, 1PT (BC)
	W	ASA2(G), 1PT (AG)
	R	ASA2(C)▲, 1PT (AC)
	W	ASB1(G), 2PT (BG)
	R	ASB1(C)▲, 2PT (BC)
1	W	ASB2(G), 2PT (AG)
	R	ASB2(C)▲, 2PT (AC)
	W	ASC1(G), 3PT (BG)
	R	ASC1(C)▲, 3PT (BC)
	W	ASC2(G), 3PT (AG)
	R	ASC2(C)▲, 3PT (AC)
		20 BLUE, WHITE & RED TWISTED PLUG WIRING (EPL)
	B	EPL(1),1PT(+20B),1PT(+20A)**
	W	EPL(2),1PT(FS2),1PT(FS1)**
	R	EPL(3),1PT(2COM),1PT(1COM)**
	B	EPL(4),2PT(+20B),2PT(+20A)**
	W	EPL(5),2PT(FS2),2PT(FS1)**
	R	EPL(6),2PT(2COM),2PT(1COM)**
	B	EPL(9),3PT(+20B),3PT(+20A)**
	W	EPL(10),3PT(FS2),3PT(FS1)**
	R	EPL(11),3PT(2COM),3PT(1COM)**
		** LAST CONNECTION POINT NOT TWISTED
1		▲ SOLDER BARE END OF THE #16 BLACK WIRE WITH THE BARE END OF THE #20 RED WIRE TO THE CATHODE (C) OF THE SCR TOGETHER AS ONE OPERATION.

REVISIONS		PRINTS TO	
3 Jan 31, 1978 SV 872 #20	1 Nov 19, 1976	5B(8)M	5J(BW)
CHG LABEL MARKING ON ASM. 1PT WAS PTA 2PT WAS PTB 3PT WAS PTC. GATE LEAD WIRES WERE PTA, PTB & PTC. TWISTED PLUG WIRING WAS PTA, PTB & PTC.	DELETED 100KVA AND G2	5R(2)BW	5(4)(1)
	2 June 23, 1977 PER AIR SS-2-1418	5L(2)BW	
	CHG'D RES MTG, ADDED "G" WIRE CONN ADDED ▲ NOTE FOR WIRE TABLE CONN	AW(BW)	
		5D(CD)	
		JA(CD)	

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department.

MADE BY Robert L. Weber Sept 23, 1976
ISSUED W. J. Ryzmanski Oct 11, 1976
APPROVALS
SPEED VARIATOR DIV OR DEPT
ERIE, PA. LOCATION
36D868890AA CONT ON SHEET SH NO.